

Computer Science & IT

Discrete Mathematics



Combinatorics

Lecture No. 01



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Recap of Previous Lecture



Topic

Practice questions



Topic

Tautology w.r.t. predicate logic



Topics to be Covered



Topic

Introduction to recurrence relation



Topic

Formation of recurrence relation



Topic

Solution of recurrence relation
using substitution method



H.W.
#Q.

Which of the following argument is/are valid?

~~A~~

$\forall x (P(x))$ follows from the premises $\exists x (P(x) \wedge Q(x))$

~~B~~

$\exists x (P(x) \wedge Q(x))$ follows from the premises $\exists x (P(x)) \wedge \exists x (Q(x))$

$\{ \exists x \{ P(x) \} \wedge \exists x \{ Q(x) \} \Rightarrow \exists x \{ P(x) \wedge Q(x) \}$ is invalid

C

$\exists x (Q(x))$ follows from the premises $\{ \forall x (P(x) \Rightarrow Q(x)), \exists y (Q(y)) \}$

~~D~~

$\sim P(a, b)$ follows from the premises $\{ \forall x \forall y \{ P(x, y) \rightarrow W(x, y) \}, \sim W(a, b) \}$

~~(C) $\exists x (P(x))$~~

$$a) \{ \exists x \{ P(x) \wedge Q(x) \} \}$$

$\Downarrow$ Existential specification

$P(a) \wedge Q(a)$ is true for some a

$\Downarrow$ simplification

$P(a)$ is true for some a

$\therefore \exists x \{ P(x) \}$ is valid / but $\forall x \{ P(x) \}$ is invalid

$$\textcircled{C} \{ \forall x \{ P(x) \rightarrow Q(x) \}, \exists y Q(y) \} \Rightarrow \exists x \{ Q(x) \}$$

$\Downarrow$ E.S.
 $Q(a)$ for some a
 $\Downarrow$ Existential generalization
 $\exists x \{ Q(x) \}$

$\therefore$ option 'C' is valid

$$\textcircled{C'} \{ \forall x \{ P(x) \rightarrow Q(x) \}, \exists y Q(y) \} \Rightarrow \exists x \{ P(x) \}$$

$\Downarrow$
 $P(a) \rightarrow Q(a)$ for all a , $Q(b)$ for some b

$\swarrow$ fallacy of affirming the consequent
 $\therefore P(a)$ for some a is invalid

$\therefore \exists x \{ P(x) \}$ need not follow.
 for $\forall x \{ P(x) \rightarrow Q(x) \}$ to be true follow.
 $P(x)$ may be false for all x

$$\textcircled{1} \{ \forall x \forall y \{ P(x,y) \rightarrow W(x,y) \} , \quad \sim W(a,b) \}$$

$\Downarrow$ Universal specification

$\Downarrow$

$$P(l,m) \rightarrow W(l,m) \text{ is true}$$

for all in domain

||| it is true for all
 $\therefore$ it will be true
 for a, b as well

$$\sim W(a,b) \text{ is true}$$

Hence

$$P(a,b) \rightarrow W(a,b) \text{ is true}$$

Modus Tollens

$$\therefore \sim P(a,b) \text{ is true}$$

{ Hence Valid
 argument }

H.W.
#Q.

Check whether the following argument is true or false?

$\forall x \{F(x) \rightarrow \sim S(x)\}$ follows from the premises

$$\{ \underbrace{\exists x(F(x) \wedge S(x))}_P \rightarrow \underbrace{\forall y(M(y) \rightarrow W(y))}_Q, \exists y(M(y) \wedge \sim W(y)) \}$$

P

Q

$$\sim [\forall y \{ \sim M(y) \vee W(y) \}]$$

$$\sim [\forall y (M(y) \rightarrow W(y))]$$

Q

$\sim Q$

M.T.

$\therefore \sim P$ is true

$$\sim [\exists x \{F(x) \wedge S(x)\}]$$

$$= \forall x \{ \sim F(x) \vee \sim S(x) \} = \forall x \{ F(x) \rightarrow \sim S(x) \}$$

Hence argument is true

Combinatorics



Topic : Recurrence Relation

Consider a sequence of real numbers $\{a_0, a_1, a_2, a_3, \dots\}$ a formula that relates $\underline{a_n}$ with one or more of the preceding terms $\{a_{n-1}, a_{n-2}, \dots\}$ is called a recurrence relation.

$$a_n = 3 \cdot a_{n-1} + a_{n-2}$$

$$a_n = 2 \cdot a_{n-1} + 2 \cdot a_{n-2} + 3 \cdot a_{n-3}$$

$$a_n = f(a_{n-1}, a_{n-2}, \dots, n)$$



Topic : Examples of recurrence relation

① Fibonacci series:

F_0 F_1 F_2 F_3 F_4 F_5 ...
 a_0 a_1 a_2 a_3 a_4 a_5 ...
1, 1, 2, 3, 5, 8, ...

$$F_n = F_{n-1} + F_{n-2}$$

where, $F_0 = 1, F_1 = 1$

② A.P. series

$$a_n = a_{n-1} + d$$

Constant { known as Common difference }

③ G.P. series

$$a_n = r \cdot a_{n-1}$$

Constant { known as Common ratio }



Topic : Linear recurrence relation

The recurrence relation of the form

$$C_0 \cdot a_n + C_1 \cdot a_{n-1} + C_2 \cdot a_{n-2} + \dots + C_k \cdot a_{n-k} = f(n) \text{ --- eqn ①}$$

is called a linear recurrence relation.

(i) In eqn ①, if $f(n) = 0$, then it is called
a homogeneous linear recurrence relation

(ii) In eqn ①, if $f(n) \neq 0$, then it is called
non-homogeneous linear recurrence relation



Topic : Formation of recurrence relation



Q.1:-> Let a_n represents the number of ways a person can climb a flight of n-steps while person is allowed to skip at most one step at a time, then
Find recurrence relation for a_n

Q.1: Soluⁿ →

In the very first move ⇒

Case ①

Either the person will not skip any step

Person does not skip any step. it can happen only in '1' way.

(And)
remaining $(n-1)$ steps can be covered in a_{n-1} ways

or

or

Case ②

the person will skip exactly '1' step.

Person skips exactly '1' step it can happen only in '1' way.

And
remaining $(n-2)$ steps can be covered in a_{n-2} ways

Case ① and Case ② are mutually Exclusive and Exhaustive

∴ The person can climb the flight of n -steps using

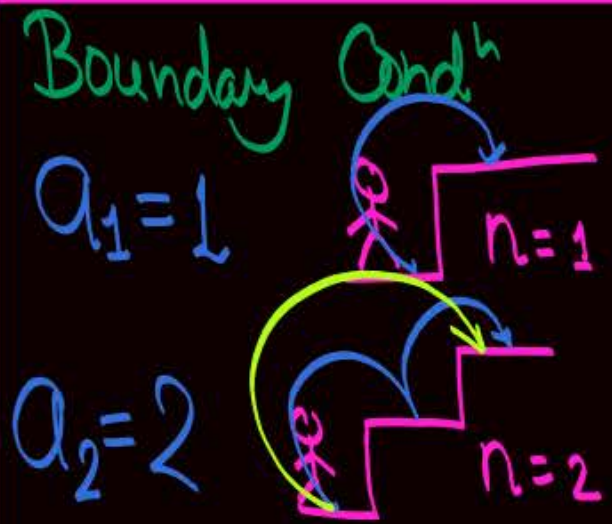
∴ No. of ways to climb n -steps

i.e. $a_n =$

$$(1 * a_{n-1}) +$$

$$(1 * a_{n-2})$$

$$a_n = a_{n-1} + a_{n-2}$$





Topic : Formation of recurrence relation

Q.2: - Let a_n represents the number of ways a person can climb a flight of n -steps while person is allowed to skip at most two steps at a time, then

ie, 0, 1 or 2
Find recurrence relation for a_n

Q.2:

Soluⁿ:-→

In the very first move ⇒

Case ①

Either the person will not skip any step

↓
Person does not skip any step. it can happen only in '1' way.

(And)

remaining $(n-1)$ steps can be covered in a_{n-1} ways

or

or

Case ②

the person will skip exactly '1' step.

↓
Person skips exactly '1' step it can happen only in '1' way.

And

remaining $(n-2)$ steps can be covered in a_{n-2} ways

Case ③

the person will skip exactly '2' step.

↓
Person skips exactly '2' steps it can happen only in '1' way.

And

remaining $(n-3)$ steps can be covered in a_{n-3} ways

∴ No. of ways to
climb a flight
of n -steps

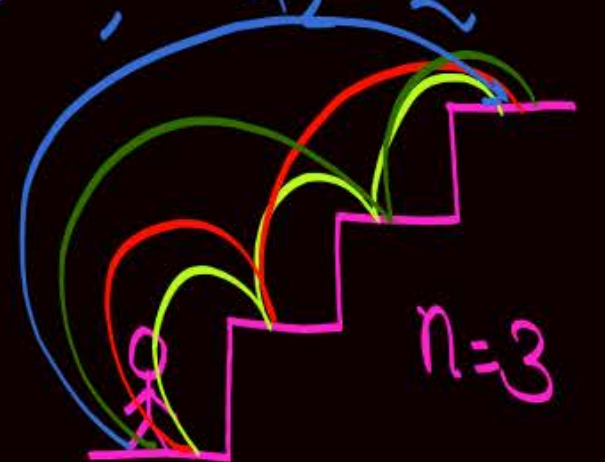
i.e. $a_n =$ Using Case ① or Using Case ② or Using Case ③

$$a_n = 1 \cdot a_{n-1} + 1 \cdot a_{n-2} + 1 \cdot a_{n-3}$$

$$a_n = a_{n-1} + a_{n-2} + a_{n-3}$$

Boundary Condⁿ : -
 $a_1 = 1$, $a_2 = 2$

$$a_3 = 4$$





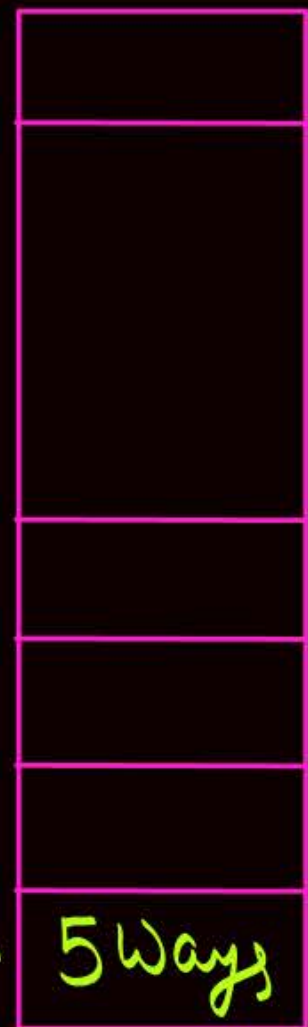
Topic : Formation of recurrence relation

Q.3:- Let a_n represents the number of ways to arrange a pile of n -chips using Red, Green, Blue, White and Gold colour chips such that no two gold colour chips are together, then
Find recurrence relation for a_n

total no.
of chips = n

A_n

initially



5 Ways

1st Chip

No restriction
on Color
Non gold

for remaining
($n-2$) chips
we have (A_{n-2}) ways

4 Ways

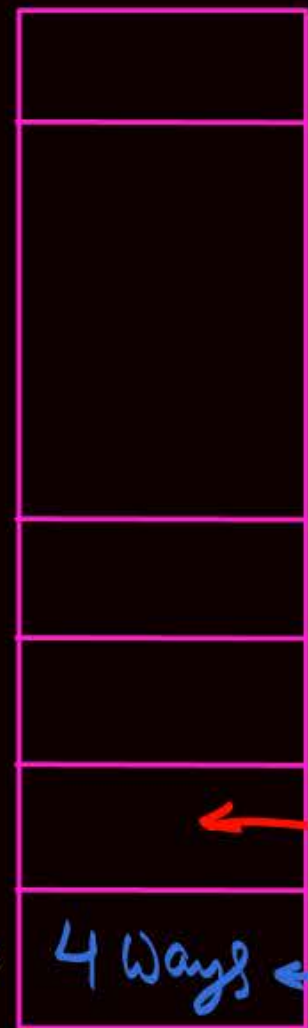
Gold

for remaining
($n-1$) chip,
(A_{n-1}) is wrong

1 Way

1st Chip

after 1st chip



4 Ways

Non gold

for remaining
($n-1$) chips
we have (A_{n-1}) ways

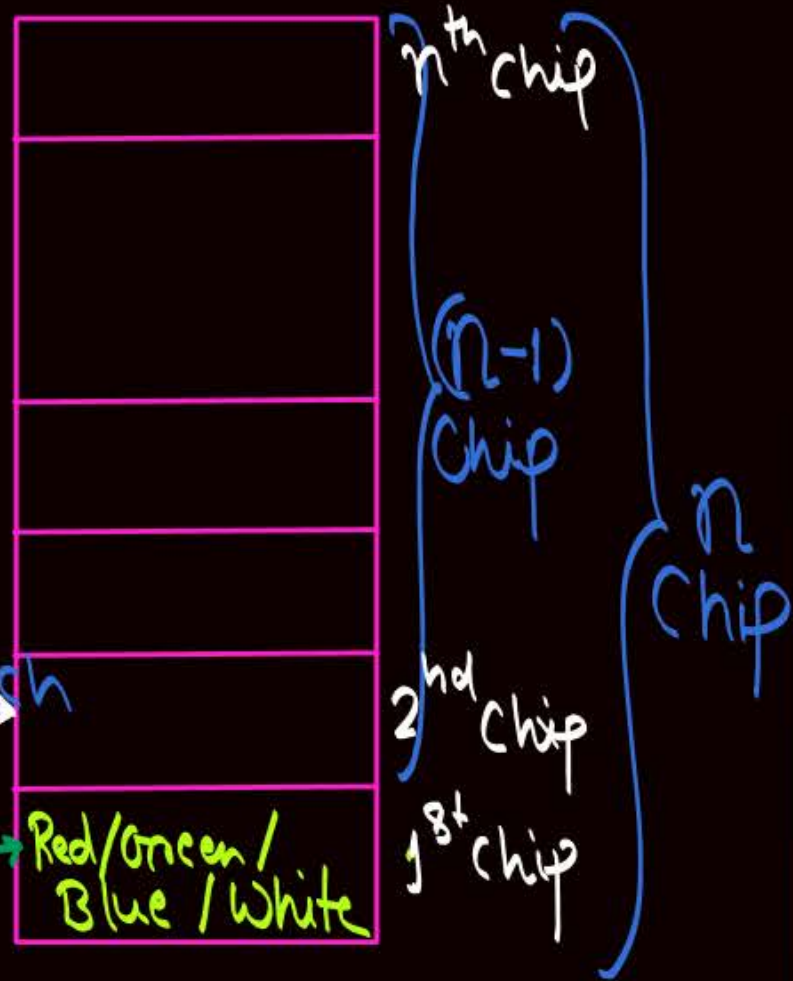
No restriction
on Color

1st Chip

Case ①: When 1st chip is of non-gold color

If 1st chip is of non-gold color then we don't have any restriction on the color of 2nd chip

i.e. from 2nd chip onwards it will be like if we are creating a pile of $(n-1)$ chip from scratch



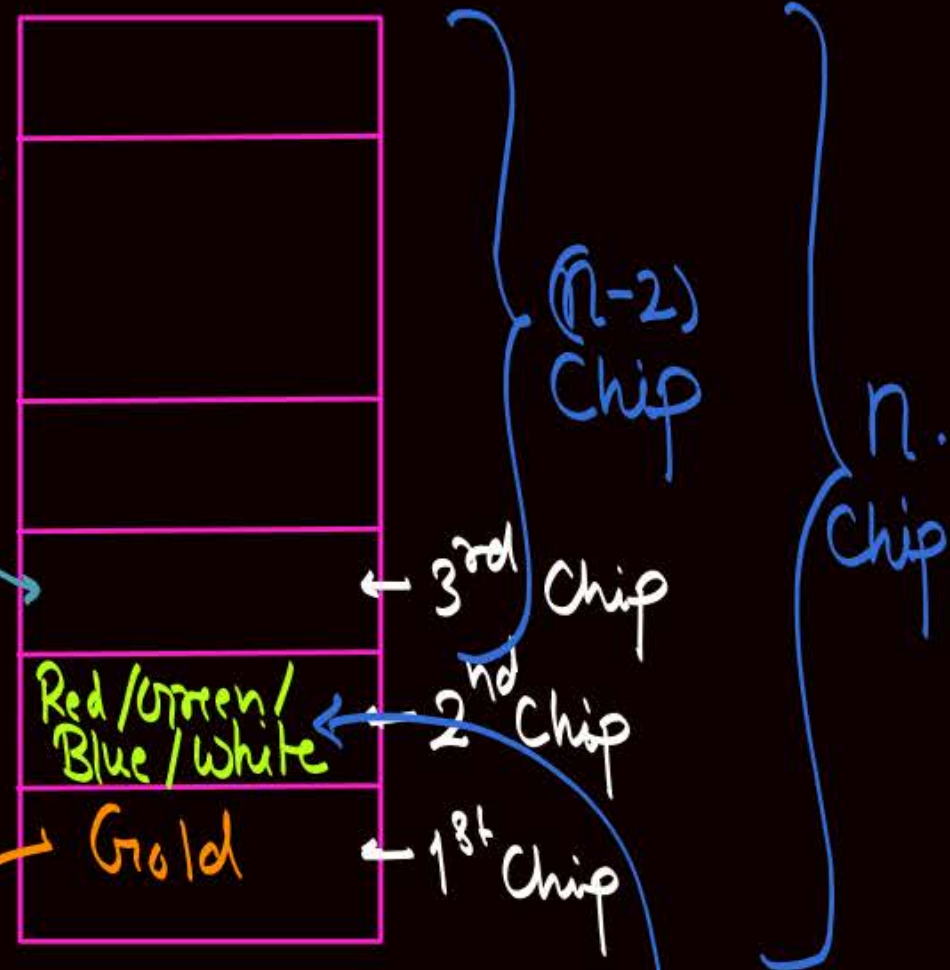
1st chip can be chosen in 4 ways

∴ No. of ways to create a pile of $(n-1)$ chip from scratch can be given by A_{n-1}

Case ②: When 1st chip is of gold color

2nd chip is of non-gold color, ∴ No restriction on color of 3rd chip

i.e. it's like if we are creating a pile of $(n-2)$ chip from scratch ∴ A_{n-2} ways



1st chip can be of gold color only in '1' way

If 1st chip is of gold color, then 2nd must be of non-gold color. ∴ 4 ways to select 2nd chip

∴ No. of ways to
Create a Pile of
n-chip.

i.e., $A_n =$

Using
Case ①

or

Using
Case ②

∴ $A_n = 4 \times A_{n-1} +$

for 1st
non-gold
Color chip

for remaining
Pile of
(n-1) chip

$1 \times 4 \times A_{n-2}$

for 1st
chip of
gold Color

and

2nd chip
can be of
non-gold
Color in
4-ways

and

for remaining
Pile of
(n-2) chip

$$A_n = 4 \cdot A_{n-1} + 4 \cdot A_{n-2}$$

5 ways

5 ways

Boundary Condⁿ

$$A_1 = 5$$

$$A_2 = (4 \times 5) + (1 \times 4) = 24 \quad \text{or} \quad A_2 = (5)^2 - 1 = 25 - 1 = 24$$



Topic : Formation of recurrence relation

H.W.

Q.

Let a_n represents the number of n -digit binary sequences of '0' and '1' with no consecutive zeros, then

Find recurrence relation for a_n



Topic : Formation of recurrence relation

H.W.
Q. Let a_n represents the number of n -digit ternary sequences of 0, 1, and 2 with even number of zeros in it, then

Find recurrence relation for a_n

Think
Carefully



2 mins Summary



✓ **Topic**

Recurrence relation

✓ **Topic**

Formation of recurrence relation

{ **Topic**

Solution of recurrence relation using substitution method

Next class

THANK - YOU